

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #8 Capacity Planning

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Agenda of the Day

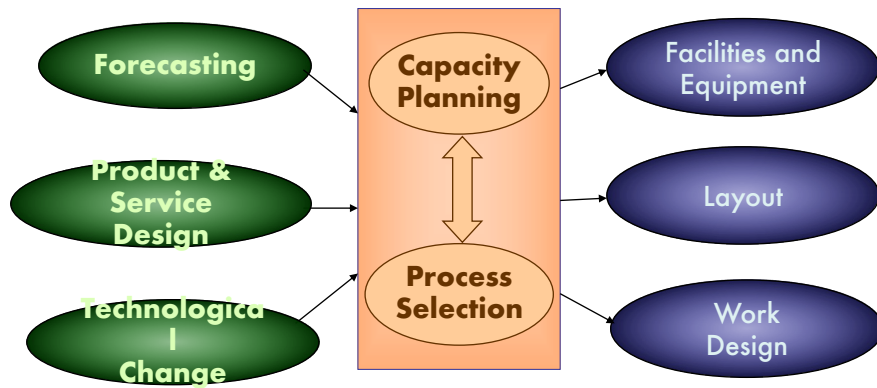
- Capacity planning
- Design vs Effective capacity
- Utilization
- Capacity planning decisions

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Process Selection and System Design



Capacity is significantly impacted by process selection and facility layout.

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Capacity Planning

- ❑ **Capacity** is the maximum output rate of a production (manufacturing or service) facility.
- ❑ **Capacity planning** is the process of establishing the output rate that may be needed at a facility to meet our forecasts.
 - Capacity is usually purchased in “chunks”
 - Strategic issues: how much and when to spend capital for additional facility & equipment
 - Tactical issues: workforce & inventory levels,
 - Operational issues: day-to-day use of equipment

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Measuring Capacity Examples

- ❑ There is no one best way to measure capacity
- ❑ **Output measures** like tables per day are easier to understand
- ❑ With multiple products, **input measures** work better

Type of Business	Input Measures of Capacity	Output Measures of Capacity
Car manufacturer	Labor hours	Cars per shift
Hospital	Available beds	Patients per month
Pizza parlor	Labor hours	Pizzas per day
Retail store	Floor space in square feet	Revenue per foot

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Capacity Information Needed

Design capacity:

- Maximum output rate under ideal conditions (maximum theoretical output)
- A bakery can make 30 custom cakes per day when pushed at holiday time

Effective capacity:

- Maximum output rate under normal (realistic) conditions (expected output)
- On the average this bakery can make 20 custom cakes per day

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Calculating Capacity Utilization

- ❑ Actual output: Rate of output actually achieved --- cannot exceed design capacity
- ❑ Utilization: Measures how much of the available capacity is actually being used:

$$\text{Utilization} = \frac{\text{actual output rate}}{\text{capacity}} (100\%)$$

- ❑ Measures effectiveness
 - Use either effective or design capacity in denominator

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Example of Operations Capacity Utilization:

In the bakery example, the design capacity is 30 custom cakes per day. Currently the bakery is producing 28 cakes per day. What is the bakery's capacity utilization relative to both design and effective capacity?

$$\text{Utilization} = \frac{\text{actual output}}{\text{design capacity}} (100\%) = \frac{28}{30} (100\%) = 93\%$$

$$\text{Efficiency} = \frac{\text{actual output}}{\text{effective capacity}} (100\%) = \frac{28}{20} (100\%) = 140\%$$

The current utilization is only slightly below its design capacity and considerably above its effective capacity

The bakery can only operate at this level for a short period of time

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Robomart

- RoboMart is a small robotic grocery warehouse that prepares online orders.
- Every hour, **online orders arrive at rate**
 $\lambda = 24 \text{ orders/hour}$.
- Each order goes through **two main stages**:
 - **Picking Stage (Station 1)**
 - Robots pick items from shelves.
 - There are **3 picking robots** working in parallel ($c_1 = 3$).
 - On average, **each robot** can complete **10 orders per hour** ($\mu_1 = 10$).
 - **Packing & Quality Check Stage (Station 2)**
 - Packed by human workers who also check items.
 - There are **2 packers** working in parallel ($c_2 = 2$).
 - On average, **each packer** can process **15 orders per hour** ($\mu_2 = 15$).
 - However, **5% of orders fail the quality check** and must be **repacked from scratch** at Station 2 (they *do not* go back to picking, only repacking).

What are the utilization rates at each station?

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Robomart

- **a) Station 1 (picking robots)**

$$\lambda_1 = \lambda = 24$$

$$c_1 = 3, \mu_1 = 10$$

$$\rho_1 = \frac{24}{3 \times 10} = \frac{24}{30} = 0.8$$

- **b) Station 2 (manual pickers)**

$$\lambda_2 = 24 + 0.05\lambda_2 \Rightarrow \lambda_2 = 25.26$$

$$c_2 = 2, \mu_2 = 15$$

$$\rho_2 = \frac{\lambda_2}{c_2 \mu_2} = \frac{25.26}{2 \times 15} = \frac{25.26}{30} \approx 0.842$$

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Robomart: Scenario-1

b) Station 2 (two pickers)

$$c_2 = 2, \mu_2 = 15$$

$$\lambda_2 = \lambda_1 + 0.12\lambda_2$$

$$\Rightarrow \lambda_2 = 2 \times 15 = 30$$

$$30 = \lambda_1 + 0.12 \times 30$$

$$\Rightarrow \lambda_1 = 26.4$$

$$\rho_2 = \frac{\lambda_2}{c_2 \mu_2} = \frac{30}{30} = 1$$

□ a) Station 1 (three robots + worker)

$$\lambda_1 = 26.4$$

$$c_1 = 3, \mu_1 = 14$$

$$\rho_1 = \frac{26.4}{3 \times 14} = \frac{26.4}{42} = 0.63$$

Overall production 26.4 per hour

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Robomart: Scenario-2

Station 1 (three robots)

$$c_1 = 3, \mu_1 = 10$$

$$\lambda_1 = \min(30, 34) = 30$$

$$\rho_1 = \frac{\lambda_1}{c_1 \times \mu_1} = \frac{30}{30} = 1$$

Station 2 (three workers)

$$\lambda_2 = 30 + 0.12\lambda_2 \Rightarrow \lambda_2 = 34.09$$

$$c_2 = 3, \mu_2 = 15$$

$$\rho_2 = \frac{\lambda_2}{c_2 \mu_2} = \frac{34.09}{3 \times 15} = \frac{34.09}{45} \approx 0.76$$

Overall production 30 per hour

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Capacity Considerations

- ❑ The **Best Operating Level** is the output that results in the lowest average unit cost
- ❑ **Economies of Scale:** If the output rate is less than the optimal level, increasing output rate results in decreasing average unit costs → the cost per unit of output drops as volume of output increases
- ❑ Reasons for economies of scale:
 - Fixed costs of buildings and equipment are spread over a larger number of units
 - Construction costs increase at a decreasing rate as facility size increases
 - Processing costs decrease due to standardization
 - Allow bulk purchasing and handling of material, decreasing the unit cost of items

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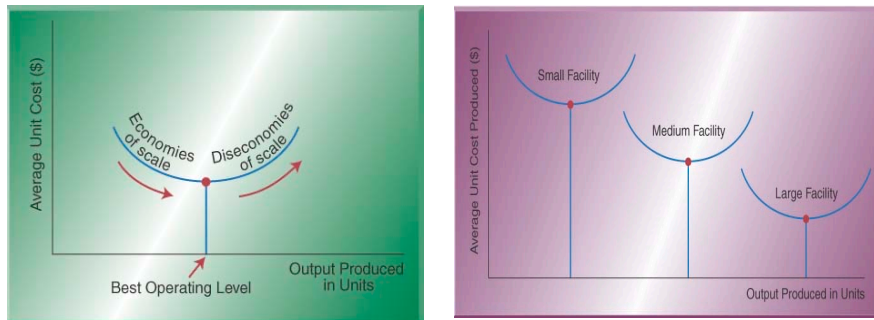
Capacity Considerations

- ❑ The **Best Operating Level** is the output that results in the lowest average unit cost
- ❑ **Diseconomies of Scale:** If the output rate is more than the optimal level, increasing the output rate results in increasing average unit costs → the cost per unit rises as volume increases
- ❑ Reasons for diseconomies of scale
 - Distribution costs increase due to traffic congestion and shipping from a centralized facility rather than multiple smaller facilities
 - Complexity increases costs
 - Inflexibility can be an issue
 - Additional levels of bureaucracy

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Best Operating Level and Size



- ❑ **Alternative 1:** Purchase one large facility, requiring one large initial investment
- ❑ **Alternative 2:** Add capacity incrementally in smaller chunks as needed

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Making Capacity Planning Decisions

- ❑ The three-step procedure for making capacity planning decisions is as follows:
 - Step 1: Identify Capacity Requirements
 - Step 2: Develop Capacity Alternatives
 - Step 3: Evaluate Capacity Alternatives

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Example

A company must choose one of three facility sizes for next year's production. What is the best configuration if the production target is 24K.

Cost Structure

- Total annual cost is:

$$C(q) = F + vq + c \left(\frac{q}{K - q} \right)$$

- F = fixed cost of the facility
- v = variable cost per unit produced
- K = maximum capacity
- c = congestion cost weight

Facility Size	Fixed Cost F	Variable Cost per Unit v	Capacity K (units/year)	Congestion Cost Weight c
Small Facility	80,000	22	12,000	4,000
Medium Facility	180,000	18	25,000	6,000
Large Facility	430,000	13	50,000	10,000

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Summary

- ❑ Capacity planning is deciding on the maximum output rate of a facility.
- ❑ Capacity planning and location decisions are often made at the same time because they are inter-related.
- ❑ To choose between capacity planning alternatives, managers may use a modeling tool like decision trees.

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